Certified bounds on the inclusive $\tau \to X_s \nu_\tau$ rate from lattice correlators with a Hausdorff moment condition on $[-1, 1]$

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Derivations and numerics prepared with Claude (Anthropic); every step is meant to be checked by the author.

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Abstract

The Chebyshev method for inclusive τ decays assumes that the lattice correlator is a positive combination of λ^t with $\lambda = \mathrm{e}^{-E} \in (0, 1]$. For domain-wall fermions with $M_5 > 1$ the transfer matrix has negative eigenvalues, and the assumption is not guaranteed. I show, line by line, how the rate is written as a linear functional of a positive measure, why the old Chebyshev coefficients leave $[-1, 1]$, and how a moment problem on $[-1, 1]$ gives *certified* upper and lower bounds on the kernel integral from the RBC/UKQCD $48^3 \times 96$ and $64^3 \times 128$ $\bar{u}s$ correlators. On these data the support $(0, 1]$ is excluded at 3σ in the $J = 0$ (time-component) channels, while $[-1, 1]$ is compatible. With the HFLAV value $\mathcal{B}_s = 0.02908(48)$ the resulting interval is $0.198 \lesssim |V_{us}| \lesssim 0.249$ (0.196–0.251 with the $\mathcal{B}_s$ error added); it is a guarantee under the stated assumptions, not a precision determination.

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1 Introduction

The inclusive hadronic width $\Gamma(\tau \to X_s \nu_\tau)$ gives $|V_{us}|$ once the spectral function of the $\bar{u}s$ currents is known. On the lattice only the Euclidean correlator $C(t)$ is available. The method of Refs. [3, 4, 5] expands the kernel in shifted Chebyshev polynomials $T_n^*(\lambda)$ of $\lambda = \mathrm{e}^{-\hat{H}}$ and uses $|\langle T_n^* \rangle| \leq 1$, which holds for a positive measure on $[0, 1]$.

Three observations motivate this note.

1. With Möbius domain-wall fermions at $M_5 = 1.8$ the one-step transfer matrix has real negative eigenvalues [1]; a positive measure on $[0, 1]$ is then not guaranteed.
2. In the old direct-sum implementation the coefficients $\langle T_n^* \rangle$ leave $[-1, 1]$ at $n \simeq 5$–8 (Sec. 6), and the resulting $|V_{us}|$ does not converge with the number of terms.
3. The kernel is singular at $E \to 0$ and is cut off by a regulator without physical meaning; a lower edge $E \geq M_K$ is physical and removes the need for it.

The remedy proposed here keeps the structure of the calculation but replaces the support $(0, 1]$ by $[-1, 1]$ and replaces the Chebyshev truncation by a moment problem whose answer is an *interval*.

1.1 The method in one page

1. **Data.** The eight γ channels of the $\bar{u}s$ correlator are combined into A_T, V_T $(J = 0)$ and A_S, V_S $(J = 1)$; the renormalised, per-volume correlator is $C_{\mathrm{ren}} = Z^2 C_{\mathrm{file}}/L^3$ (Eq. 6) and the rate is a linear functional of it, $\Sigma_{\mathrm{tot}} = \sum_c \int \mathrm{d}\hat{E}\, (2/\hat{E}) K_J \hat{\rho}_c$ with $\Gamma_{\mathrm{app}} = (a^{-1})^2 \hat{\Sigma}$ (Eqs. 7–11).
2. **Hypothesis (new).** $C(t) = \int \mathrm{d}\sigma(\lambda)\, \lambda^{t-1}$ for $t \geq 1$ with $\sigma \geq 0$ supported on $[-1, \mathrm{e}^{-M_0}]$, where M_0 is the kaon mass (Eq. 13). The old method has σ on $(0, \mathrm{e}^{-M_0}]$ only.
3. **Constraints.** The moments $m_k = \int \mathrm{d}\sigma \lambda^k$ must lie in $C(k+1) \pm z\delta_{k+1}$, $k = 0, \ldots, T-1$ (Eq. 15).
4. **Bounds.** The largest and smallest value of $\int g\, \mathrm{d}\sigma$ over all such σ is a linear program; its dual is a polynomial $p \geq g$ that certifies the bound (Sec. 4). Negative-λ states can absorb moment budget but never contribute to the rate.
5. **Rate.** $|V_{us}| = \sqrt{\mathcal{B}_s \Gamma_\tau / (S_{EW} \frac{G_F^2 m_\tau^3}{16\pi} \Sigma_{\mathrm{tot}})}$, with $\mathcal{B}_s = 0.02908(48)$ from HFLAV 2023 (Sec. 11.6.1 of arXiv:2411.18639).

Result on the RBC/UKQCD $48^3 \times 96$ and $64^3 \times 128$ data. The support $(0, 1]$ cannot reproduce the $J = 0$ channels within 3σ (it needs $z^* = 3.3$–19); $[-1, 1]$ can ($z^* \leq 0.1$). The certified interval is $0.198 \leq |V_{us}| \leq 0.249$ (both ensembles, $z = 3$, $T = 18$), or $0.196 \leq |V_{us}| \leq 0.251$ with the $\mathcal{B}_s$ error added; it contains the HFLAV lattice-inclusive value 0.2189(19) and the OPE value 0.2184(21). It is a guarantee, about $\pm 11\%$ wide, not a precision determination.

Conventions. E is an energy, $\lambda = \mathrm{e}^{-E}$. Lattice quantities carry no unit (a hat when ambiguity is possible); a^{-1} converts to GeV. T is the number of correlator values used, $t = 1, \ldots, T$.

2 From the decay rate to a linear functional of the correlator

2.1 The rate

Following your note (Eq. (28)), with $y = Q^2/m_\tau^2$ and $\Gamma_0 = m_\tau^5 G_F^2/(192\pi^3)$,

$$R \equiv \frac{\Gamma}{\Gamma_0} = 12\pi^2 |V_{us}|^2 S_{EW} \int_0^{m_\tau^2} \frac{\mathrm{d}Q^2}{m_\tau^2} (1-y)^2 \Big[(1+2y)\rho^{(1)}(Q^2) + \rho^{(0)}(Q^2) \Big]. \tag{1}$$

Multiplying by Γ_0 and using $12\pi^2/(192\pi^3) = 1/(16\pi)$,

$$\Gamma(\tau \to X_s \nu) = |V_{us}|^2 \, S_{EW} \, \frac{G_F^2 m_\tau^3}{16\pi} \, \Gamma_{\rm app}, \tag{2}$$

$$\Gamma_{\rm app} \equiv \int_0^{m_\tau^2} \mathrm{d}Q^2 \Big[K_1(y)\rho^{(1)}(Q^2) + K_0(y)\rho^{(0)}(Q^2) \Big], \tag{3}$$

$$K_1(y) = (1-y)^2(1+2y)\,\theta(1-y), \qquad K_0(y) = (1-y)^2\,\theta(1-y). \tag{4}$$

This is Eq. (44) of your note with $\Gamma_{\rm Exp} \to \mathcal{B}_s \Gamma_\tau$ in the numerator.

Check with the kaon pole. Take $\rho^{(0)} = f_K^2 \delta(Q^2 - M_K^2)$ (your Eq. (25)). Then $\Gamma_{\rm app} = f_K^2 K_0(M_K^2/m_\tau^2) = f_K^2(1 - M_K^2/m_\tau^2)^2$, which reproduces Eq. (26), $\Gamma(\tau \to K\nu) = G_F^2 f_K^2 m_\tau^3 (1 - M_K^2/m_\tau^2)^2 |V_{us}|^2/(16\pi)$. The K^* with $\rho^{(1)} = f_{K^*}^2 \delta(Q^2 - M_{K^*}^2)$ gives $f_{K^*}^2(1-y)^2(1+2y)$, Eq. (27).

2.2 Spectral representation of the correlator

Your Eq. (29) is

$$C(t) = \int_0^\infty \mathrm{d}Q \, Q^2 \rho(Q^2) \, \mathrm{e}^{-Qt} \;\equiv\; \int_0^\infty \mathrm{d}E \, \rho_E(E) \, \mathrm{e}^{-Et}, \qquad \rho_E(E) = E^2 \rho(E^2). \tag{5}$$

The overall normalisation is fixed by the one-state case, Eqs. (48) and (51) of your note: a single state of mass M and decay constant f gives the renormalised, per-volume correlator

$$C_{\rm ren}(t) \equiv \frac{Z^2 \, C_{\rm file}(t)}{L^3} = \frac{M f^2}{2} \, \mathrm{e}^{-Mt}, \qquad f = Z \sqrt{\frac{2A}{ML^3}} \;\; (C_{\rm file} \simeq A \mathrm{e}^{-Mt}), \tag{6}$$

i.e. $\rho_E = \frac{1}{2} M f^2 \delta(E - M)$. With $\delta(E - M) = 2M\,\delta(E^2 - M^2)$ this is $\rho(Q^2) = \rho_E/E^2 = f^2 \delta(Q^2 - M^2)$, consistent with Eq. (25). The general case follows by linearity. Here $Z = Z_A = Z_V$ (your Tables 2–4).

2.3 Rewriting $\Gamma_{\rm app}$ in the energy variable

Substitute $Q^2 = E^2$, $\mathrm{d}Q^2 = 2E\,\mathrm{d}E$ in $\Gamma_{\rm app}$ and use $\rho(E^2) = \rho_E(E)/E^2$:

$$\int \mathrm{d}Q^2 K(Q^2)\rho(Q^2) = \int \mathrm{d}E \, 2E \, K \, \frac{\rho_E(E)}{E^2} = \int_0^\infty \mathrm{d}E \; \frac{2}{E} \, K(E) \, \rho_E(E). \tag{7}$$

Check: $\rho_E = \frac{1}{2} M f^2 \delta(E - M)$ gives $\frac{2}{M} \frac{1}{2} M f^2 K = f^2 K$, as it must.

2.4 Lattice units

Let $\hat{E} = aE$ and let $c(t) = Z^2 C_{\rm file}/L^3$ be the correlator in lattice units, $c(t) = \int \mathrm{d}\hat{E}\, \hat{\rho}(\hat{E}) \mathrm{e}^{-\hat{E}t}$. For one state $\hat{M} = aM$ and $\hat{f} = af$ in Eq. (6). Then, with $\hat{\Sigma} \equiv \int \mathrm{d}\hat{E}\, (2/\hat{E}) K \, \hat{\rho}$,

$$\hat{\Sigma} = \hat{f}^2 K = a^2 f^2 K = a^2 \, \Gamma_{\rm app} \qquad \Longrightarrow \qquad \Gamma_{\rm app} = (a^{-1})^2 \, \hat{\Sigma}. \tag{8}$$

By linearity this holds for any $\hat{\rho}$. *It is* $(a^{-1})^2$, *not* $(a^{-1})^3$; K is evaluated at $y = (a^{-1}\hat{E})^2/m_\tau^2$, so the kernel edge sits at $\hat{E} = m_\tau a = 1.027$ (48I) and 0.753 (64I).

2.5 Which component carries $J = 0$ and which $J = 1$

Write the vacuum polarisation as $\Pi^{\mu\nu}(q) = (q^\mu q^\nu - q^2 g^{\mu\nu})\Pi^{(1)} + q^\mu q^\nu \Pi^{(0)}$. At zero spatial momentum $q = (q^0, \vec{0})$:

$$\Pi^{00} = (q_0^2 - q^2)\Pi^{(1)} + q_0^2 \Pi^{(0)} = q_0^2\, \Pi^{(0)} \qquad (q^2 = q_0^2), \tag{9}$$

$$\Pi^{ii} = -q^2 g^{ii}\, \Pi^{(1)} = q^2\, \Pi^{(1)} \qquad (\text{no sum over } i,\ g^{ii} = -1). \tag{10}$$

Hence the temporal components ($A_T = \gamma_5\gamma_4$, $V_T = \gamma_4$) carry $J = 0$ and the spatial ones, averaged over i, (A_S, V_S) carry $J = 1$:

$$\Sigma_{\text{tot}} = \sum_{X\in\{A,V\}} \left[\Sigma^{(0)}_{X_T} + \Sigma^{(1)}_{X_S}\right], \qquad \Sigma_c^{(J)} = (a^{-1})^2 \int \mathrm{d}\hat{E}\, \frac{2}{\hat{E}} K_J\, \hat{\rho}_c. \tag{11}$$

In the RBC files the eight gamma matrices are $g = 1, 2, 4 \to V_{x,y,z}$, $8 \to V_T$, $14, 13, 11 \to A_{x,y,z}$, $7 \to A_T$. *Remark.* The old `vus()` assigns $J = 1$ to both axial terms and $J = 0$ to both vector terms, which disagrees with Eq. (11) and with the mock code in the same file.

3 The support of the spectral measure

3.1 Positive transfer matrix

If the one-step transfer matrix $\hat{T} = \mathrm{e}^{-\hat{H}}$ is Hermitian and positive, with eigenvalues $\lambda_n = \mathrm{e}^{-E_n} \in (0, 1]$ and overlaps $c_n = \langle n|J|0\rangle$,

$$C(t) = \sum_n |c_n|^2 \lambda_n^t = \int \mathrm{d}\nu(\lambda)\, \lambda^t, \qquad \nu \geq 0, \quad \operatorname{supp}\nu \subset (0, 1]. \tag{12}$$

This is the hypothesis behind the shifted-Chebyshev method.

3.2 $M_5 > 1$

Syritsyn and Negele [1] construct $\hat{T}$ for Shamir domain-wall fermions: its eigenvalues are real, and negative ones appear for $M_5 > 1$. They do not discuss the sign of the weights. If only link reflection positivity holds, the natural form is

$$C(t) = \sum_n |c_n|^2 \lambda_n^{t-1} = \int \mathrm{d}\sigma(\lambda)\, \lambda^{t-1}, \qquad t \geq 1, \qquad \sigma = \sum_n |c_n|^2 \delta_{\lambda_n} \geq 0, \quad \operatorname{supp}\sigma \subset [-1, 1]. \tag{13}$$

Define the moments $m_k \equiv \int \mathrm{d}\sigma\, \lambda^k = C(k+1)$, $k = 0, 1, \ldots$ ($C(0)$ is dropped: contact terms). For $\lambda > 0$ the relation to the spectral density is $\mathrm{d}\sigma = \lambda\hat{\rho}\, \mathrm{d}\hat{E}$, because $\int \hat{\rho}\, \mathrm{e}^{-\hat{E}t} \mathrm{d}\hat{E} = \int \lambda^{t-1} (\lambda\hat{\rho}\, \mathrm{d}\hat{E})$. States with $\lambda < 0$ carry $\sigma \geq 0$ in this convention (a negative weight in $C(t) = \int \lambda^t\, \mathrm{d}\nu$ with $\mathrm{d}\nu = \mathrm{d}\sigma/\lambda$).

Hypotheses. $H_{[0,1]}$: $\operatorname{supp}\sigma \subset (0, 1]$ (old). $H_{[-1,1]}$: $\operatorname{supp}\sigma \subset [-1, 1]$ (new). $H_{[-1,1]}$ is weaker, contains $H_{[0,1]}$, and is *not proven* for $M_5 = 1.8$ Möbius fermions; it can only be tested.

Parity of the start time. For the sequence $s_k = C(t_s + k) = \int \mathrm{d}\sigma\, \lambda^{t_s-1}\lambda^k$ to be moments of a positive measure on $[-1, 1]$ we need $\lambda^{t_s-1} \geq 0$ there, i.e. $t_s - 1$ even, t_s odd. For even t_s the negative-λ states enter with negative weight. The old method normalises with $2t_0 = 1$, an odd t_s.

4 Certified bounds: the new method

4.1 The problem

Per channel c with kernel J, let $\kappa_J(\lambda) = \frac{2}{E} K_J((a^{-1}E)^2/m_\tau^2)$ with $E = -\ln\lambda$. From Eqs. (11) and $\mathrm{d}\sigma = \lambda\hat{\rho}\,\mathrm{d}\hat{E}$,

$$\hat{\Sigma}_c = \int \mathrm{d}\sigma(\lambda)\, g(\lambda), \qquad g(\lambda) = \begin{cases} \kappa_J(\lambda)/\lambda, & 0 < \lambda \le \mathrm{e}^{-M_0} \\ 0, & \lambda < 0. \end{cases} \tag{14}$$

States with $\lambda < 0$ are lattice artefacts: they contribute nothing to the physical rate but they consume the moment budget. The lower edge $E \ge M_0$ (the kaon, the lightest $\bar{u}s$ state) is necessary because $g \sim 2/E$ diverges at $E \to 0$. Bounds are sought for all positive measures consistent with the data:

$$\hat{\Sigma}^{\pm} = \sup / \inf \int g\,\mathrm{d}\sigma \quad \text{s.t.} \quad \sigma \ge 0 \text{ on } S, \quad |m_k - C(k+1)| \le z\,\delta_{k+1}, \quad k = 0, \dots, T-1, \tag{15}$$

with $S = [-1, \mathrm{e}^{-M_0}]$ (new) or $(0, \mathrm{e}^{-M_0}]$ (old), and δ_t the jackknife error of $C(t)$.

4.2 Weak duality (the certificate)

Proposition 1. *Let $p(\lambda) = \sum_{k=0}^{T-1} p_k \lambda^k$ satisfy $p \ge g$ on S. Then for every σ feasible in (15)*

$$\int g\,\mathrm{d}\sigma \;\le\; \sum_k p_k C(k+1) + z \sum_k |p_k|\,\delta_{k+1}. \tag{16}$$

Proof. $\int g\,\mathrm{d}\sigma \le \int p\,\mathrm{d}\sigma = \sum_k p_k m_k$ (since $\sigma \ge 0$ and $p \ge g$). Write $m_k = C(k+1) + \epsilon_k$ with $|\epsilon_k| \le z\delta_{k+1}$; then $\sum p_k \epsilon_k \le z \sum |p_k| \delta_{k+1}$. □

The lower bound is the mirror image with a polynomial $q \le g$ and $-z\sum |q_k|\delta_{k+1}$. Minimising the right-hand side over p is the dual of (15); strong duality holds for the moment problem on a compact set, and for polynomial g the condition $p - g \ge 0$ on $[-1, 1]$ is equivalent to $p - g = s_0 + (1 - \lambda^2)s_1$ with sums of squares s_0, s_1 (Lukács), i.e. a semidefinite program.

4.3 Numerics and what is rigorous

g is not a polynomial, so I solve the primal on a fine grid of λ (about 6×10^3 points, variables $u_i \ge 0$ rescaled by $C(1)$, `scipy.optimize.linprog`/HiGHS). This gives an *inner* approximation. The dual variables $y_k^{\pm}$ of the two inequality rows of moment k give $p_k = (y_k^+ - y_k^-)/\delta_{k+1}$ with $p \ge g$ on the grid. *The rigorous step:* $p - g$ is evaluated on a dense grid of 2×10^5 points of the whole support, p is lifted by the largest violation Δ ($p_0 \to p_0 + \Delta$), and the proposition is applied. On the data the certified and the grid bounds differ by at most $3 \times 10^{-5}\,\mathrm{GeV}^2$ (Table 3). (The check on a dense grid is numerical, not an analytic proof of $p \ge g$; g is smooth and the lift is small.)

Statistical meaning. Each box is $\pm z\delta_t$. With Gaussian errors the T boxes jointly contain the truth with probability at least $1 - T \cdot 0.0027$ for $z = 3$ (Bonferroni): $\ge 95\%$ for $T = 18$. The covariance between t is not used.

Calibration. A single state ($m = 0.4937\,\mathrm{GeV}$, $f = 0.1564\,\mathrm{GeV}$ on 48I, errors 10^{-3}) gives $\Sigma \in [0.02075, 0.02089]\,\mathrm{GeV}^2$ for both supports, around the exact $f^2(1 - M^2/m_\tau^2)^2 = 0.02083$; the certificate reproduces the grid bound to 5 digits.

4.4 Feasibility

The LP minimising $\max_t |\sum_i w_i \lambda_i^{t-1} - C(t)|/\delta_t$ gives the smallest z^* at which a positive measure on S reproduces the data. At $z = 3$ the old support is therefore *infeasible* for A_T and V_T, at both

z^* at $T = 14$ (old $[0,1]$ / new $[-1,1]$)	A_T	V_T	A_S	V_S
48I	3.28 / 0.09	5.22 / 0.07	0.64 / 0.39	0.27 / 0.20
64I	9.06 / 0.06	18.72 / 0.08	0.75 / 0.10	0.78 / 0.05

Table 1: Minimal tolerance (in σ) for the data to be reproduced by a positive measure. For $T \geq 6$ the $J = 0$ channels need $z^* > 3$ with the old support.

lattice spacings and for every $T \geq 6$, which is the moment-problem version of "the Chebyshev coefficients exceed 1".

5 Testing the hypothesis: Hausdorff positivity

5.1 Necessary conditions, proved

Let $P(\lambda) = \sum_{i=0}^{n} x_i \lambda^i$ be a real polynomial and $s_k = \int \mathrm{d}\sigma\, \lambda^k\, \lambda^{t_s - 1}$. If $\sigma \geq 0$ on $[-1, 1]$:

$$x^\top H x = \sum_{ij} x_i x_j\, s_{i+j} = \int \mathrm{d}\sigma\, \lambda^{t_s-1} P(\lambda)^2 \;\geq 0, \qquad H_{ij} = s_{i+j}, \tag{17}$$

$$x^\top H^{\mathrm{loc}} x = \int \mathrm{d}\sigma\, \lambda^{t_s-1}(1-\lambda^2) P(\lambda)^2 \;\geq 0, \qquad H^{\mathrm{loc}}_{ij} = s_{i+j} - s_{i+j+2}. \tag{18}$$

For support $[0, 1]$ the second matrix is $H^{\mathrm{loc}}_{ij} = s_{i+j+1} - s_{i+j+2}$, from the weight $\lambda(1-\lambda)$. Schmüdgen [2] proves that for $s_0, \ldots, s_{2n}$ these conditions are also *sufficient* (no rank condition on a compact interval). We only need necessity: a significantly negative eigenvalue *excludes* the hypothesis; passing means compatibility, not proof.

5.2 Scaling and statistics

Scaling $H \to D^{-1} H D^{-1}$ with $D = \mathrm{diag}(\sqrt{|H_{ii}|})$ is a congruence and does not change positive semidefiniteness (Sylvester's law), and removes the strong decay of s_k. For each (t_s, n) we compute the smallest eigenvalue $\mu_{\min}$ and its jackknife error over configurations, and call a matrix a *violation* if $\mu_{\min}/\sigma < -3$, a *pass* if > 3. (The rows of the data files carry no configuration label; the autocorrelation along the row order is at the noise level, $|\mathrm{ac}| < 0.05$ for lags 1–100, so the jackknife error does not depend on the assumed ordering.)

5.3 Result

The $J = 0$ channels violate $[0, 1]$ at odd t_s and $[-1, 1]$ at even t_s, as predicted by Eq. (13). The $J = 1$ channels do not discriminate. Because the same data enter many (t_s, n) tests, isolated 3σ violations are expected by chance; V_T is also the least clean channel (a non-conserved local current for $m_u \neq m_s$).

channel	support	odd t_s (pass/inc/viol)	even t_s
V_S, A_S ($J=1$), 48I and 64I	$[-1,1]$	5–7 / 1–3 / 0	4–6 / 2–4 / 0
V_T ($J=0$), 64I	$[-1,1]$	5/3/0	2/3/**3**
V_T ($J=0$), 64I	$[0,1]$	0/5/**3**	3/5/0
A_T ($J=0$), 64I	$[-1,1]$	5/3/0	3/3/2
A_T ($J=0$), 64I	$[0,1]$	3/3/**2**	3/5/0

Table 2: Localizing matrices, $n \geq 2$, $t_s \leq 8$. At odd t_s the support $[-1,1]$ is never violated in any channel on either ensemble; the $J=0$ channels show the pattern expected for negative-λ states. Full table: `hankel_scan.py`.

6 The Chebyshev coefficients

6.1 Definitions and the old relation

With $\lambda = (1+\cos\theta)/2 = \cos^2(\theta/2)$ set $T_j^*(\lambda) = \cos j\theta = T_j(2\lambda-1)$. Normalise $\bar{C}(t) = C(t+2t_0)/C(2t_0) = m_t/m_0 \equiv \langle\lambda^t\rangle$. Then

$$\lambda^t = \cos^{2t}\tfrac{\theta}{2} = 2^{-2t}\sum_{k=0}^{2t}\binom{2t}{k}\mathrm{e}^{\mathrm{i}(t-k)\theta} = 2^{1-2t}\Big[\tfrac{1}{2}\binom{2t}{t} + \sum_{j=1}^{t}\binom{2t}{t-j}\cos j\theta\Big], \tag{19}$$

$$\bar{C}(t) = 2^{1-2t}\Big[\tfrac{1}{2}\binom{2t}{t} + \sum_{j=1}^{t}\binom{2t}{t-j}\langle T_j^*\rangle\Big], \tag{20}$$

which is your Eq. (56) (with $\langle T_0^*\rangle = 1$ by normalisation). The old method fits $\langle T_j^*\rangle$ to $\bar{C}(t)$, $t = 1,\ldots,T$, with the prior $|\langle T_j^*\rangle| < 1$ (through erfinv).

6.2 The bound, and when it fails

Lemma 1. *If $\sigma \geq 0$ is normalised on $[0,1]$, $|\langle T_n^*\rangle| \leq 1$. If $\sigma \geq 0$ is normalised on $[-1,1]$, $|\langle T_n\rangle| \leq 1$.*

Proof. $|T_n^*| \leq 1$ on $[0,1]$ and $|T_n| \leq 1$ on $[-1,1]$, so $|\langle T\rangle| \leq \int |T|\,\mathrm{d}\sigma \leq \int \mathrm{d}\sigma = 1$. □

Now let a weight w sit at $\lambda = -\varepsilon$ ($\varepsilon > 0$). Since $2\lambda - 1 = -1 - 2\varepsilon < -1$,

$$T_n^*(-\varepsilon) = (-1)^n \cosh\big(n\,\mathrm{arccosh}(1+2\varepsilon)\big) \simeq \tfrac{1}{2}(-1)^n \mathrm{e}^{n\kappa}, \quad \kappa = \mathrm{arccosh}(1+2\varepsilon), \tag{21}$$

so the contribution $w\,T_n^*(-\varepsilon)$ exceeds 1 for $n \gtrsim \ln(2/w)/\kappa$. For $\varepsilon = 0.1$ ($\kappa = 0.62$): $n \gtrsim 8.5$ for $w = 1\%$ and $n \gtrsim 4.8$ for $w = 10\%$. A small negative-λ component is therefore enough to push the old coefficients out of $[-1,1]$ at the n observed in the data.

6.3 Noise amplification

The coefficients of $T_n^*(x)$ alternate in sign (all roots lie in $(0,1)$, Descartes), so $\sum_k |c_k| = |T_n^*(-1)| = T_n(3) \simeq \frac{1}{2}(3+2\sqrt{2})^n$. The coefficients of $T_n(x)$ alternate in x^2, so $\sum_k |c_k| = |T_n(\mathrm{i})| \simeq \frac{1}{2}(1+\sqrt{2})^n$. At $n = 12$ these are 7.7×10^8 and 2.0×10^4: the $[-1,1]$ basis amplifies the statistical noise of the moments $\sim 4\times10^4$ times less. The price is a coarser resolution in λ (about a factor 2 in the polynomial degree).

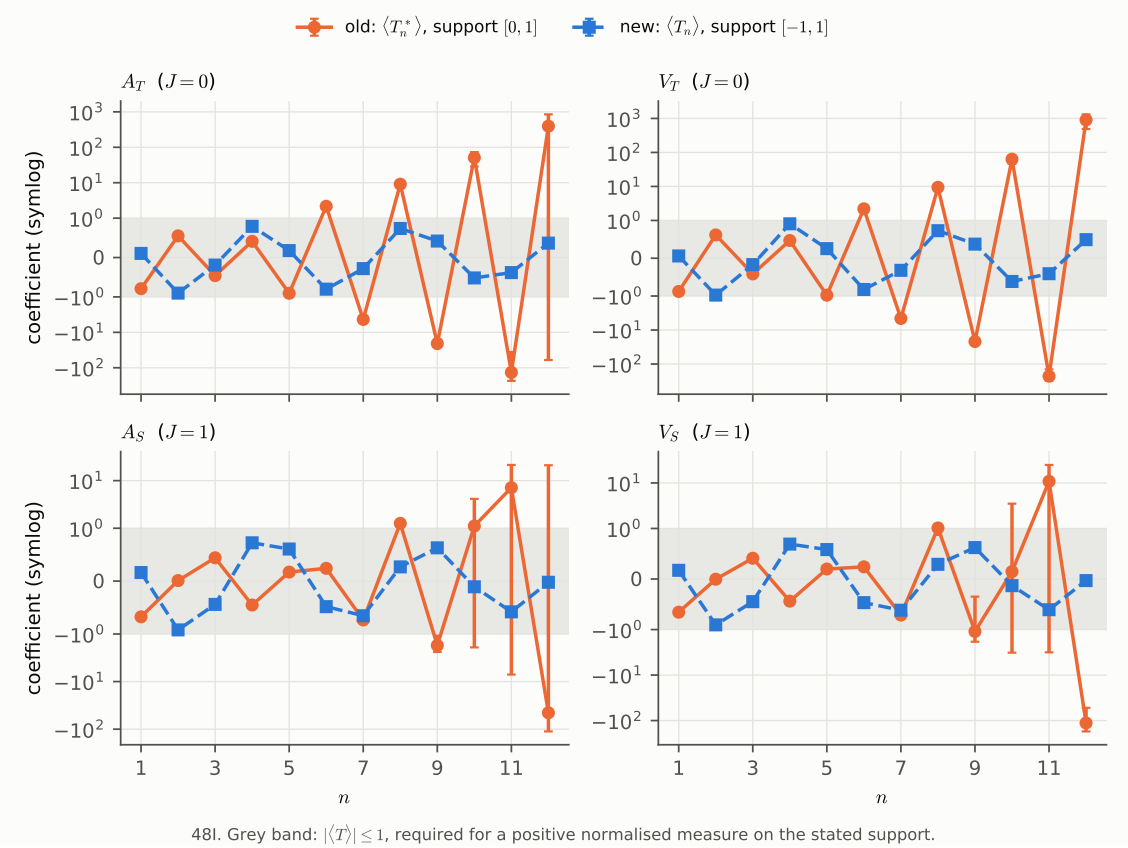


Figure 1: $\langle T_n^* \rangle$ (old, orange) and $\langle T_n \rangle$ (new, blue) on 48I, four channels, jackknife errors; symmetric-log axis; the grey band is $[-1, 1]$. The old coefficients leave the band at $n = 5$–8 (e.g. A_T, $n = 6$: 2.16 ± 0.04) and reach 10^2–10^3 at $n = 12$. The new ones stay inside for all channels and $n \leq 12$ (maximum 0.98, error ≤ 0.02). They follow $\cos(n\pi/2)$ because the measure sits near $\lambda = 0$, so this is a necessary, weak test.

6.4 Data

7 Results

7.1 Channel bounds

$\hat{\Sigma}$ bounds [GeV^2], $T = 18$, $z = 3$, certified	A_T	V_T	A_S	V_S
48I new $[-1, 1]$	[0.02048, 0.02119]	[0.00039, 0.00135]	[0.01021, 0.02894]	[0.03764, 0.059
64I new $[-1, 1]$	[0.02048, 0.02125]	[0.00038, 0.00127]	[0.00617, 0.02761]	[0.03653, 0.058
old $[0, 1]$	infeasible		feasible, slightly tighter lower bou	

Table 3: Certified bounds per channel. The kaon pole alone gives $f_K^2(1 - M_K^2/m_\tau^2)^2 = 0.02084$, inside the A_T interval: the normalisation of Sec. 2 is confirmed on real data. Certified and grid bounds differ by at most 3×10^{-5} GeV^2.

7.2 $|V_{us}|$

Eq. (44) with $\Gamma_{\text{Exp}} = \mathcal{B}_s \Gamma_\tau$ gives

$$|V_{us}| = \sqrt{\frac{\mathcal{B}_s \, \Gamma_\tau}{S_{EW} \, \dfrac{G_F^2 m_\tau^3}{16\pi} \, \Sigma_{\text{tot}}}}, \qquad \Sigma_{\text{tot}} = \sum_c \Sigma_c^{\pm}, \tag{22}$$

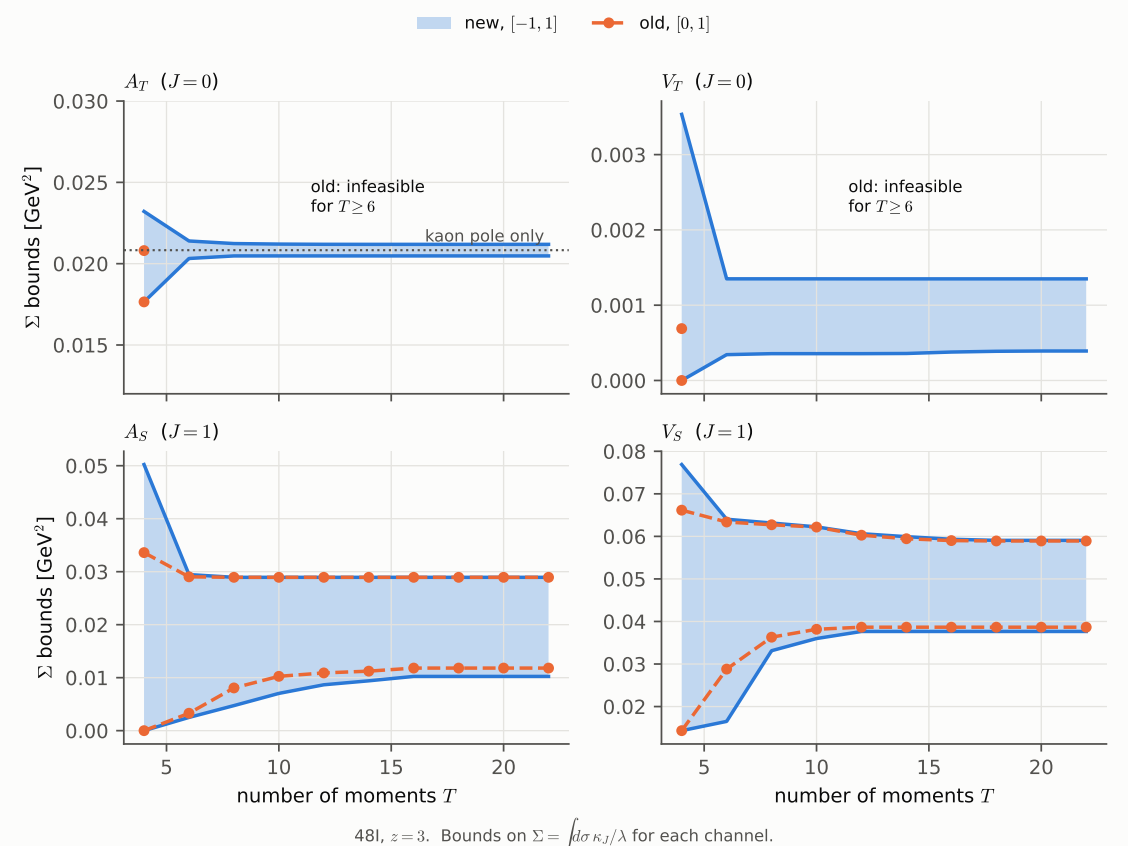


Figure 2: Bounds on $\hat{\Sigma}$ versus the number of moments T (48I, $z = 3$). New support: blue band. Old support: orange dashed, where feasible; infeasible for A_T, V_T from $T = 6$. Beyond $T \approx 18$ the relative error of $C(t)$ exceeds 25% and the bounds no longer improve. The $J = 1$ axial channel is the loosest.

with $\mathcal{B}_s = 0.02908 \pm 0.00048$ (HFLAV 2023, arXiv:2411.18639, Sec. 11.6.1, Eq. (246); the fit table gives $\mathcal{B}(\tau^- \to X_s^- \nu_\tau) = 0.0291(5)$), $\Gamma_\tau = \hbar/\tau_\tau = 2.267 \times 10^{-12}\,\mathrm{GeV}$, $S_{EW} = 1.0201$, $G_F^2 m_\tau^3/(16\pi) = 1.518 \times 10^{-11}\,\mathrm{GeV}^{-1}$. Bounds of the four independent channels add, which is conservative. Then $|V_{us}| \in [\sqrt{c/\Sigma^+_{\mathrm{tot}}}, \sqrt{c/\Sigma^-_{\mathrm{tot}}}]$ with $c = \mathcal{B}_s \Gamma_\tau/(S_{EW} G_F^2 m_\tau^3/16\pi)$.

ensemble	support	Σ_{tot} [GeV^2]	$\lvert V_{us}\rvert$	width
48I	old $[0,1]$	infeasible	—	—
64I	old $[0,1]$	infeasible	—	—
48I	new $[-1,1]$	[0.0687, 0.1105]	[0.1962, 0.2489]	11.8%
64I	new $[-1,1]$	[0.0636, 0.1089]	[0.1977, 0.2588]	13.4%
both ensembles, overlap ($z = 3$)			**[0.1977, 0.2488]**	
with the $\mathcal{B}_s$ error ($\pm 1\sigma$) added			[0.1961, 0.2509]	
both ensembles, overlap ($z = 2$)			[0.1998, 0.2351]	

Table 4: $T = 18$, $\mathcal{B}_s = 0.02908(48)$; $|V_{us}| \propto \sqrt{\mathcal{B}_s}$, so the $\mathcal{B}_s$ error shifts the interval ends by $\mp 0.8\%$. The 48I and 64I intervals are not combined into one fit. The value $\Sigma_{\mathrm{tot}} = 0.0846\,\mathrm{GeV}^2$ that corresponds to $|V_{us}| = 0.2243$ with this $\mathcal{B}_s$ lies inside both intervals. For context (not used as input), HFLAV 2023 quotes $|V_{us}| = 0.2184(21)$ (OPE) and 0.2189(19) (lattice inclusive, Eq. (249)); both are inside the interval.

7.3 Comparison with the old direct fit

Figure 3 shows the certified interval versus T together with the values from the original code (run unchanged except for removing `cupy`), for both kernel assignments of Sec. 2.5. The old values

change by ±30% between consecutive T and their quoted errors ($\approx 6 \times 10^{-4}$) do not include the truncation error. Points to check in the old code (details in `OLD_CODE_NOTES.md`): the kernel

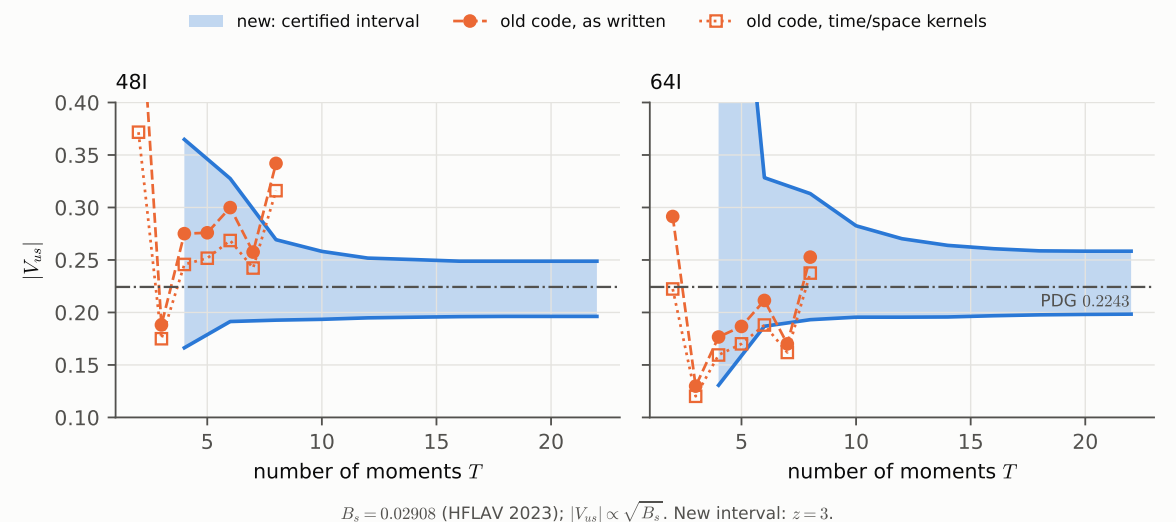


Figure 3: $|V_{us}|$ from the original code (orange; filled: as written, open: time/space kernels) and the certified interval (blue), versus T. $\mathcal{B}_s = 0.02908$ (HFLAV 2023).

assignment, the ignored `-t0` (the argument lands in `ND`), the absence of $\mathcal{B}_s$ in the $|V_{us}|$ formula, and the position of Z_A^2 (it should multiply Σ; in the code it divides inside the square root).

8 What is and is not shown

- The $(0, 1]$ support is *excluded* by the $J = 0$ data at $\geq 3.3\sigma$ (48I) and $\geq 9\sigma$ (64I); $[-1, 1]$ is compatible. This does not depend on any $|V_{us}|$ number.
- The interval is wide ($\approx \pm 11\%$). It is a guarantee, not a competitive determination; the $J = 1$ axial channel limits it.
- It is valid only if (i) $\sigma \geq 0$ on $[-1, 1]$ holds for this action (tested at odd t_s, not proven), (ii) the lowest state is the kaon, (iii) $\mathcal{B}_s$ (HFLAV 2023), Z, a^{-1}, M_0 are correct, (iv) the Gaussian error model holds. With $z = 3$ and $T = 18$ the statistical confidence is $\geq 95\%$.
- Not included: continuum extrapolation (one lattice spacing each), the errors of a^{-1} ($\approx 0.2\%$ on $|V_{us}|$) and Z, the full covariance (boxes are used), lattice artefacts of the local currents, finite-volume effects beyond the positivity of the measure.
- Next steps: an exact free-field Möbius check at $M_5 = 1.8$ (a single negative pivot would be decisive); the experimental $\mathcal{B}_s$; a SOS/SDP implementation with the exact (non-grid) polynomial certificate; the full covariance as an ellipsoid constraint.

A Reproducing the numbers

All code is in `hausdorff/` of the repository; run from the directory that contains `RBCQCD/`.

`hankel_scan.py`	Hausdorff positivity scan (Table 2)
`cheb_report.py`	$\langle T_n^* \rangle$ and $\langle T_n \rangle$ (Fig. 1)
`bounds_report.py`, `bounds_T.py`	feasibility z^* and bounds versus T
`vus_interval.py`, `vus_certified.py`	channel sums, certificate check, $\lvert V_{us} \rvert$ interval (`BS=...` sets $\mathcal{B}_s$)
`safe_vus.py`	library and single-state calibration (`python safe_vus.py`)
`old_code_check/`	the original `AllClass.py` run without `cupy`
`paper/make_figs.py`	figures

Parameters: 48I: $a^{-1} = 1.7295\,\mathrm{GeV}$, $Z = 0.71076$, $M_0 = 0.28816$, $L = 48$, $N_t = 96$, 88 configurations $\times$ 48 sources; 64I: $a^{-1} = 2.3586\,\mathrm{GeV}$, $Z = 0.74293$, $M_0 = 0.21539$, $L = 64$, $N_t = 128$, 40×64 (the split is inferred from the row count; the Hankel scan gives identical verdicts for 40×64 and 80×32).

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